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FINAL PROJECT

COMPARISON OF BUSINESS MODELS ON STARBUCKS' QUARTERLY REVENUES

FROM 2009 TO 2025

LECTURER: Prof. Mariangela Guidolin

STUDENTS: Diana Andrade (2141895), Mahsa Ekrahi (2141873), Kristina Koleva (2144044)

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Introduction

Starbucks is one of the world's most recognizable coffeehouse brands, whose global expansion, strong customer loyalty, distinctive store atmosphere, and emphasis on customer experience have contributed to its rising popularity, especially among younger generations. To understand the dynamics behind its financial performance, this project analyzes **Starbucks' quarterly revenue from 2009 to 2025**, examining how consumer behavior, market conditions, and major global events shape its revenue patterns. Using a range of time-series approaches—including classical models, ARIMA and seasonal methods, as well as innovation-diffusion models such as BM, GBM, and GGM—the study aims to forecast future revenue trajectories, while also considering advertising expenditure as an external factor influencing growth.

Problem description

Identifying the most appropriate model for a specific dataset can become challenging due to different factors. In this study case, Starbucks' revenues experienced a sharp decline in 2020, likely as a result of the COVID-19 pandemic. This behavior deviates substantially from both prior and subsequent observations, breaking the continuity of the increasing trend and yearly seasonality of the time series. For this reason, it is necessary to identify a model that can properly represent both the typical trend of the series and the anomalous observations associated with the pandemic period.

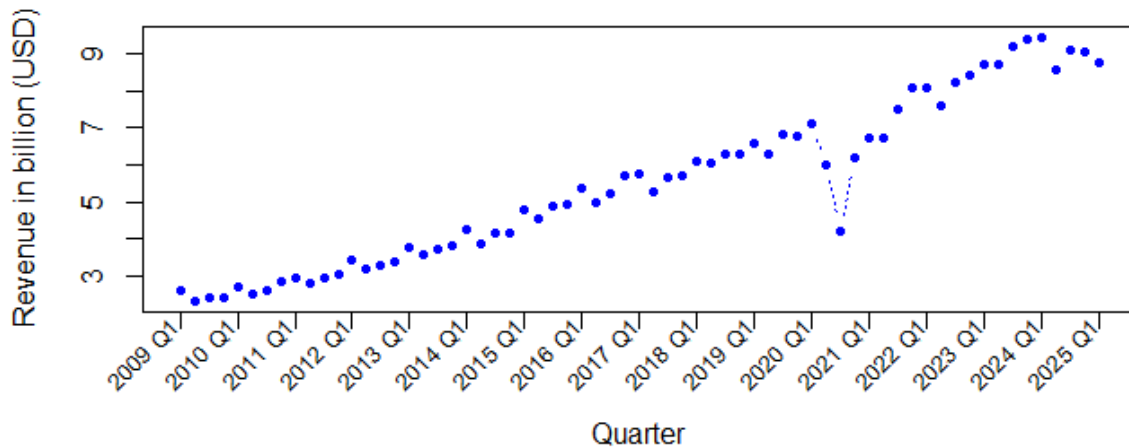
Data presentation

During the development of this project 4 datasets were implemented, 2 qualitative and 2 quantitative. All datasets were obtained from [Statista Website](#), and more specific information and plots of these can be found in the following lines.

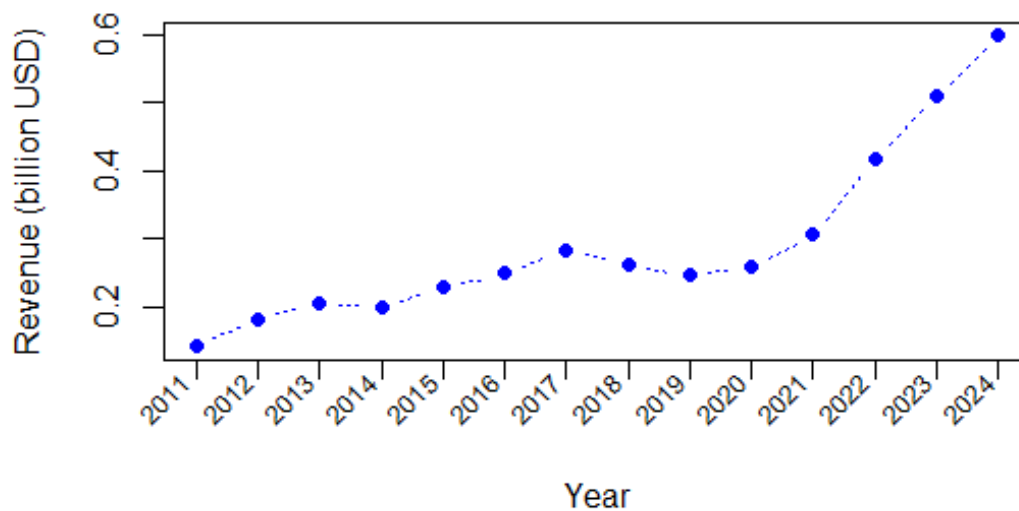
- **Quantitative datasets:**
 - a. **Starbucks' revenues (Starbucks, & U.S. Securities and Exchange Commission, 2025):** Starbucks' quarterly revenue from 2009 to 2025, there are 65 data points depicting the revenue in billion dollars (USD). There are 4 data points per year, but the year of 2025 contains only the first quarter. In *Graph 1*, we can appreciate the seasonality and increasing trend of the data, as well as the dip in 2020.
 - b. **Starbucks' advertising spendings (Starbucks, 2025):** Starbucks' yearly advertising spendings from 2011 to 2024. The dataset contains 14 data points, each of them corresponding to one year, and the amounts are in millions of dollars (USD). In *Graph 2*, we can appreciate the dataset's increasing trend.
- **Qualitative datasets:**
 - a. **Starbucks' positive aspects in Italy (MyGoodMoods, 2016):** This dataset provides an overview of the opinion of consumers about Starbucks' positive aspects in Italy in 2016. According to the data, the majority of respondents highly valued the possibility to access Wi-Fi (18%) and the general atmosphere (17%). More arguments can be found in *Graph 3*.

- b. **Reasons to like Starbucks in Japan (LINE Research, 2023):** Dataset of a survey conducted in June 2023 in Japan. It was found that the seasonal and new types of menus offered by Starbucks was the most common reason the coffee chain was liked by its Japanese clientele. The rest of the reasons can be found in *Graph 4*.

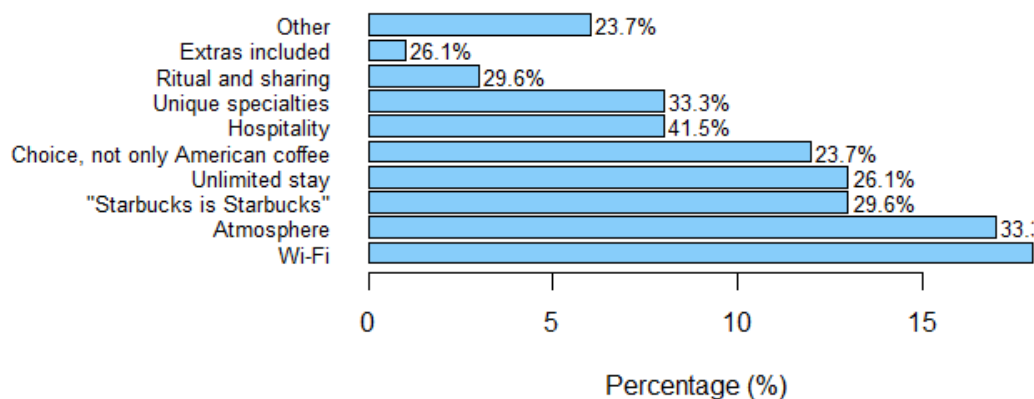
Graph 1. Starbucks' quarterly revenues from 2009 to 2025



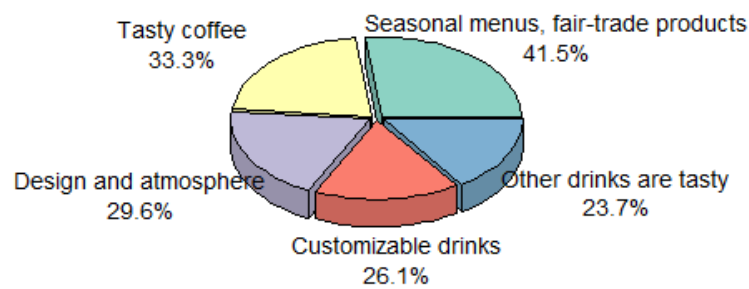
Graph 2. Starbucks' yearly advertising spendings from 2011 to 2024



Graph 3. Survey results: Positive aspects about Starbucks in Italy



Graph 4. Reasons to like Starbucks in Japan



Modeling hypothesis

In order to define our first hypothesis, it was necessary to perform a deeper analysis of the Starbucks' revenues dataset behaviour. This analysis was mainly performed by interpreting the ACF and PACF graphs (*Figure 1*).

The autocorrelation function (ACF) exhibits many significant autocorrelations across multiple lags, with values decaying slowly, indicating that the series is non-stationary due to an underlying trend. The partial autocorrelation function (PACF) shows a dominant spike at lag 1 followed by a rapid decline, which is characteristic of a first-order autoregressive relationship.

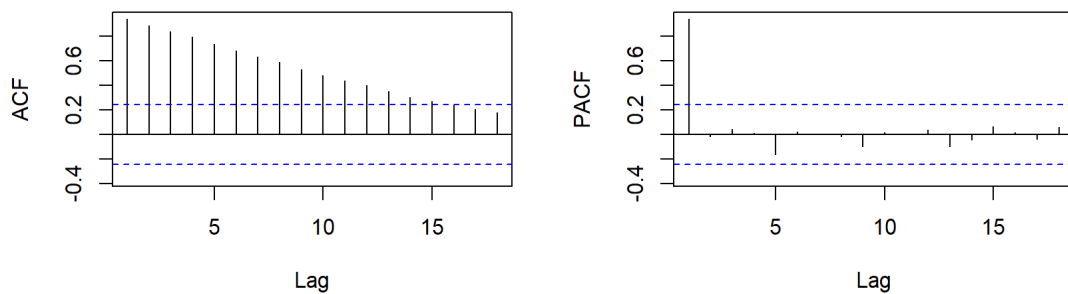


Figure 1. ACF and PACF of Starbucks' revenues from 2009 to 2025

Based on this information and the details of the dataset mentioned previously, the following hypotheses were created:

1. A linear regression model will only fit the trend and seasonality properly
2. A GBM with exponential shock is required to fit the dip in the 2020 year
3. ARIMA models will provide the best solution for a dataset with autoregressive relationship as this one.
4. GAM models will outstand ARIMA models' performance due to their flexibility to interpret non linear trends and Prophet models could handle the dip better, thanks to the holidays effects.
5. Advertising expenses can help to justify Starbucks' growth and the rapid recovery in revenues after the pandemic

Main research question

- Which model provided the best solution for the Starbucks' revenues dataset?
- Increasing the expenses of advertising is actually helping to increase the revenues?

- Fitting the dip presented in 2020 properly is enough to obtain a good model?

Models implemented

Our goal is to find the best mathematical model to represent the dataset. We start from the simplest models and adjust the complexity of these models, based on their performance. We evaluated the performance of our models according to the Akaike Information Criterion (AIC), so from now on every time we consider that a model is better than another, we conclude this according to this value. Since it's not possible to obtain the AIC for non linear regression models, we also computed the MSE to monitor the performance of these models.

Implementation on quarterly data:

- A. **Linear regression:** Time series
 - a. This is the simplest model to implement and the one where we expected the worst results. However, trend and seasonality components are included in order to improve the performance of the model.
- B. **Non-linear regression models:** BM, GBM, GGM
 - a. First, we used the BM model to estimate the parameters p , q and m that later we implemented on the GBM model. The GBM model was fitted using first a rectangular and then an exponential shock. Even though both models provided a considerably high R^2 , the one with the exponential shock obtained a lower residual error and in consequence a better performance.
 - b. A GGM was also implemented with the purpose of comparison, but no significant results were obtained.
- C. **ARIMA and SARIMA:** We fitted 4 Arima models manually and also one model was implemented through automatic parameter selection using `auto.arima`. Due to the existence of seasonality in our data, the SARIMA models were also fitted. In the following section of *Results* we will present the AIC of each of these models.
- D. **Short-term forecasting:** We fitted the simple exponential smoothing, the Holt's with also the damped version, the Holt-Winters and finally the ETS in order to have an automated forecasting method selected on the lowest AIC.
- E. **Prophet model:** Different prophet models were fitted, given its ability to work with seasonality and significant events, considering linear and logistic growth, additive and multiplicative seasonality. Specifically in this implementation we also tried to handle the shock as a holiday which has yielded good results.

Implementation on yearly data:

- A. **ARIMAX:** In order to justify the rapid recovery of Starbucks' revenues, we decided to use the advertising expenses as an external variable. However, this data was presented yearly and from 2011 to 2024. Since our data was originally presented quarterly, it was necessary to compute the yearly revenues of Starbucks from the previous period specified. Another important detail to take care of is the scale, since the Starbucks' revenues are provided in billion USD, while the advertising spendings in million USD. Considering that 1,000 million corresponds to 1 billion, we'll simply divide our advertising dataset by 1000 in order to match our original scale in billion (USD).

- B. **Generalized Additive model and Prophet model:** Taking into account the spendings advertising dataset and the time, the initial try was the linear model as a special case of GAM, which reveals to be better than the GAM with the smoothing on the spendings as also the non significance of the non parametric effect shows. Later we considered the prophet model with linear and logistic growth, to make a comparison between the best GAM model and the best prophet model.

Results

QUARTERLY MODELS

A. Linear vs non linear regression models

As we mentioned before, our linear model is a time series with trend and seasonality, while our non linear models include the BM, GBM with exponential shock and GGM. We will use MSE to compare them since it's not possible to compute the AIC for some models. In *Figure 2*, it can be observed how all the models fitted on the actual data. From a visual perspective, it's evident that the time series was the one who better captured the seasonality effect, while the GBM with exponential shock was the only one with the ability to capture the decline presented in 2020.

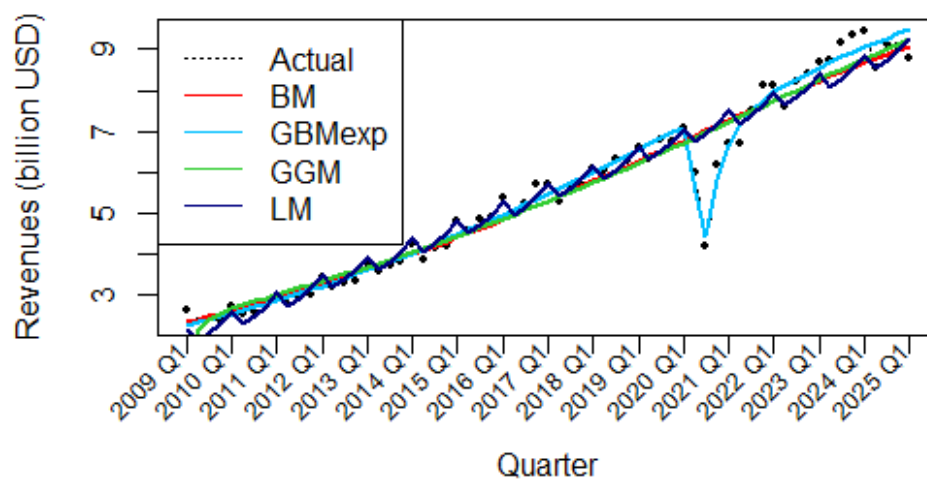


Figure 2. Comparison between models with linear and non linear regression (LM, BM, GBMexp and GGM)

From a mathematical point of view, these results can be confirmed by evaluating the MSE values on *Table 1*, where the GBMexp obtained the lowest MSE with a value of *0.1203*, followed by the one with rectangular shock with *0.1375* and finally the time series with *0.2264*.

For simplicity purposes we will only review the ACF and residuals of the time series and GBMexp representing the best linear and non linear regression models, respectively.

Table 1. MSE comparison between linear and non linear regression models	
MODEL	MSE
LM	0.2264
BM	2.0596
GBM rect	0.1375
GBMexp	0.1203
GGM	1.9014

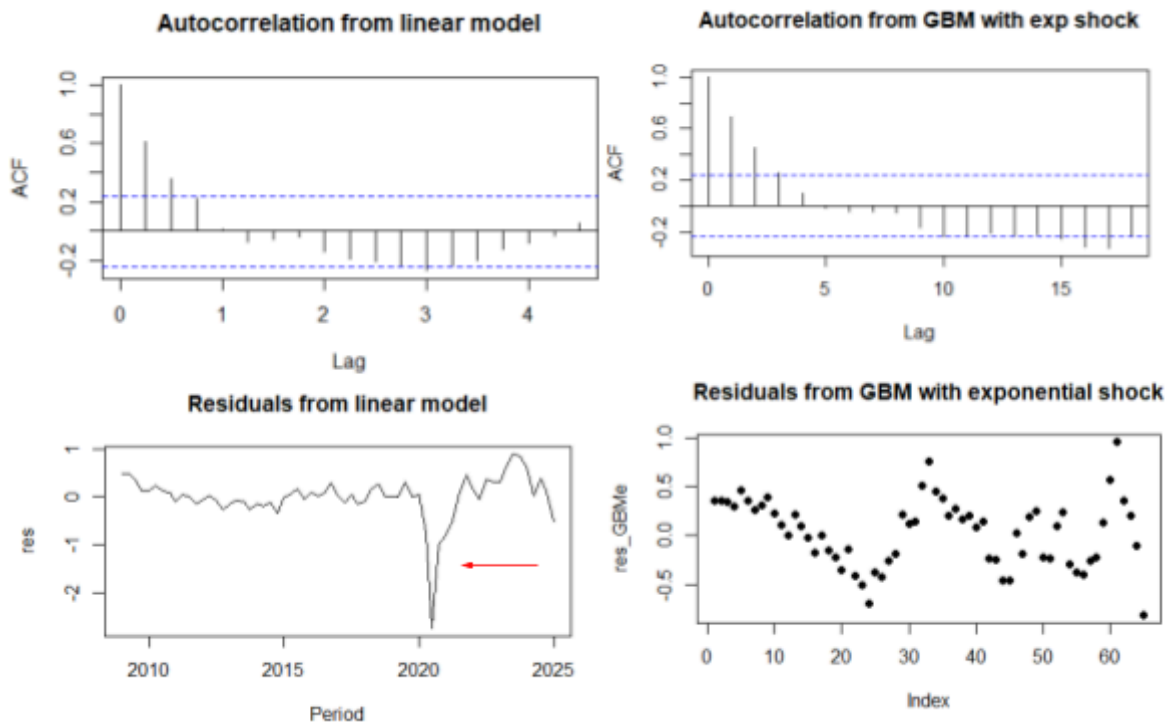


Figure 3. Plots of ACF and residuals of the time series and GBM with exponential shock models

As observed in the *Figure 3*, even though both models provided considerably good results in terms of MSE values, their ACF plots present the first three lags significant and decreasing monotonically. This is a typical behavior of a stationary autoregressive (AR) structure, which suggests that the residuals are not fully white noise. In fact, this can be confirmed by observing the residuals plot where we can observe a dip in the linear model residuals and a sort of sinusoidal wave shape for the GBMexp. Consequently, this is a strong indication that some features remained unmodeled, and it suggests the incorporation of a differencing component may be needed to reduce trend and seasonality (Guidolin, 2025, 4)

B. ARIMA and SARIMA

The ARIMA models allow us to control the AR, differencing and Moving Average (MA) component of the model. Based on our results, it's evident that the AR component had no influence on the model's performance. In fact, the biggest improvement was obtained by adding a first level of differencing, allowing the model to go from an AIC of 225.1214 to 93.905. In the following experiments we can observe that increasing the differencing and AR components does not affect this value considerably. Actually, the best model was obtained using the auto.arima tool, which provided an ARIMA(0,1,2), meaning first

level of differencing and a MA component of second order, which basically confirms that current shocks are affected by the previous two periods, in this case quarters.

Additionally, we have the SARIMA model that offers us the advantage to control the (p,d,q) parameters of the seasonal part independently from the non seasonal component (Guidolin, 2025, 28). In our case, the SARIMA (1,1,1)(1,1,1)[4] provided even better results than our previous ARIMA model, with an AIC of 82.737. For simplicity purposes, we will only analyze

deeper the ACF and residuals of these two models.

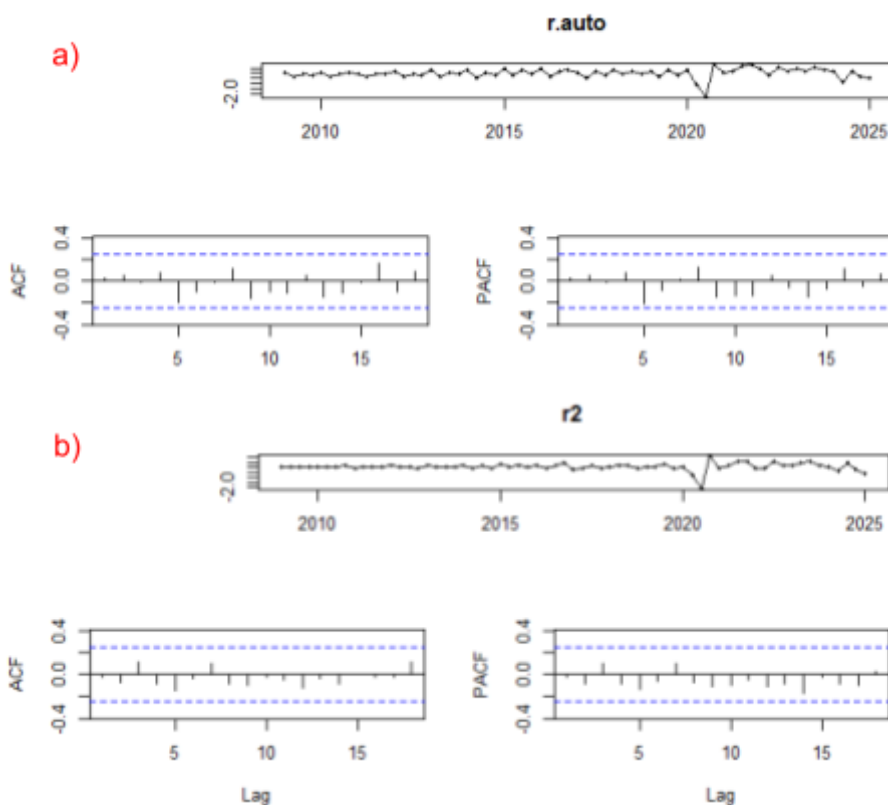


Figure 4. Residuals, ACF and PACF plots of a) ARIMA (0,1,2), and b) SARIMA(1,1,1)(1,1,1)[4]

As observed in Figure 4, both models provided considerably good results in terms of the ACF and PACF plots, since none of the lags they contained were significant. However, the residuals plot is still not fully convincing since the ARIMA is still showing some pattern, probably related to the seasonality, this feature is considerably improved in the SARIMA model; however, the dip in

2020 is a feature that none of our implementations have managed to model completely.

C. Short-term forecasting

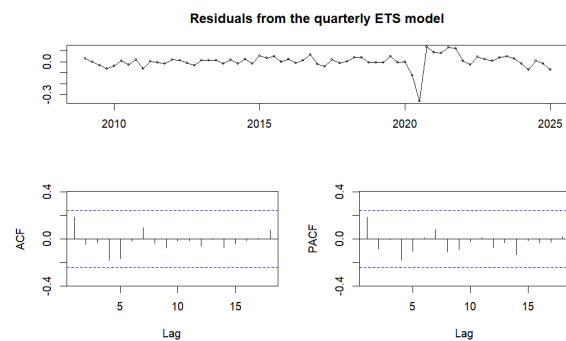
Among the many implemented short-term forecasting methods, the Holt-Winters in this case is the first that can handle both trend and seasonality. Moreover two modalities of it were observed: additive and multiplicative. While in the additive the fit is smoother, in the second one the captured behavior is jumpy and follows better the pattern at the beginning of the series, but despite this difference the forecast is almost the same and in terms of ACF and PACF they both show a good fit.

Additionally the automated ETS model, specifically the one with multiplicative error, additive trend and additive seasonality, was implemented through which we reached the best AIC value, at least for the short-term forecasting. As observed in Figure 5, again there are no significant lags, but the pattern of the residuals is still recognizable.

Table 3. AIC comparison across exponential smoothing methods

MODEL	AIC
Simple exp smooth	181.4967
Holt	179.9052
Damped Holt	185.4953
Additive Holt-Winters	171.2720
Multiplicative Holt-Winters	155.1605
ETS	142.3077

Figure 5. Residuals for the ETS models



D. Prophet model

In this section it's evident that the multiplicative linear and logistic prophet models have a lower MSE. At first glance, the multiplicative linear captures better the datapoints, the residuals are centered around zero with the exception of the shock period, where a large residual is observed with mostly non significant spikes in the ACF.

The multiplicative logistic, in contrast, has an evident stronger negative autocorrelation at the initial and final lags, suggesting a remaining structure in the residuals with a sinusoidal pattern.

On the other side, for the additive linear and logistic prophet we have a higher MSE and in both a strong lag-1 autocorrelation, with also similar residuals.

Table 4. MSE comparison across prophet models			
SEASONALITY	GROWTH	MSE	HOLIDAY
Additive	Linear	0.1553	
	Logistic	0.1707	
	Linear	0.1458	Lockdown
Multiplicative	Linear	0.0615	
	Logistic	0.1474	
	Linear	0.0418	Lockdown

In all of these trials, the dip in the 2020 is never adequately captured as the residuals plot explains but in order to improve the performance, an interesting way is to handle the shock as a holiday. We decided to implement just linear growth, as previous Prophet models showed that linear growth fits better. Since the assumption is that the drop in the dataset is due to the covid-19 pandemic, it's possible to delineate as holidays the periods affected by the lockdowns. In this way for each lockdown it's readable the start date, the end date

and for how many days the effect will last. By not specifying any future dates, the lockdowns are intended as non-recurring, meaning that lockdowns will not recur in the future dataframe. Again, as before the multiplicative has a lower MSE but because of the significant residual autocorrelation and of the overfitting, the additive is preferred due to the white-noise, addressing an issue that other Prophet models weren't able to solve.

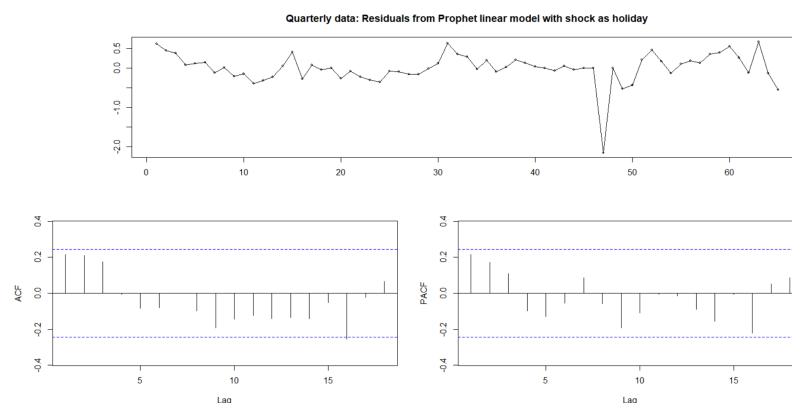


Figure 6. Residuals for the prophet model with linear growth

YEARLY MODELS

A. ARIMAX

ARIMAX is a version of the ARIMA model that allows us to include an external variable as a regressive parameter, which is optimal to explain and justify our series behaviour. For this example, we used a yearly dataset of Starbucks' advertising spendings from 2011 to 2024.

Table 5. Comparison of AIC for ARIMAX's models	
MODEL	AIC
ARIMAX (1,0,1)	63.93328
ARIMAX (1,1,1)	71.39082
ARIMAX (0,1,0)	58.59716

For this experiment we will focus on the model with the best performance which is the auto.ARIMAX(0,1,0). In order to determine if the advertising spendings are significant for our model we have to do a t-test. Considering the coefficient and standard error provided by the summary, we have that $t = \frac{\beta}{s.e.} = \frac{12.28}{12.77} = 0.96$. Since $0.96 > 0.05$, we reject the hypothesis that the advertising spendings influence on the revenues.

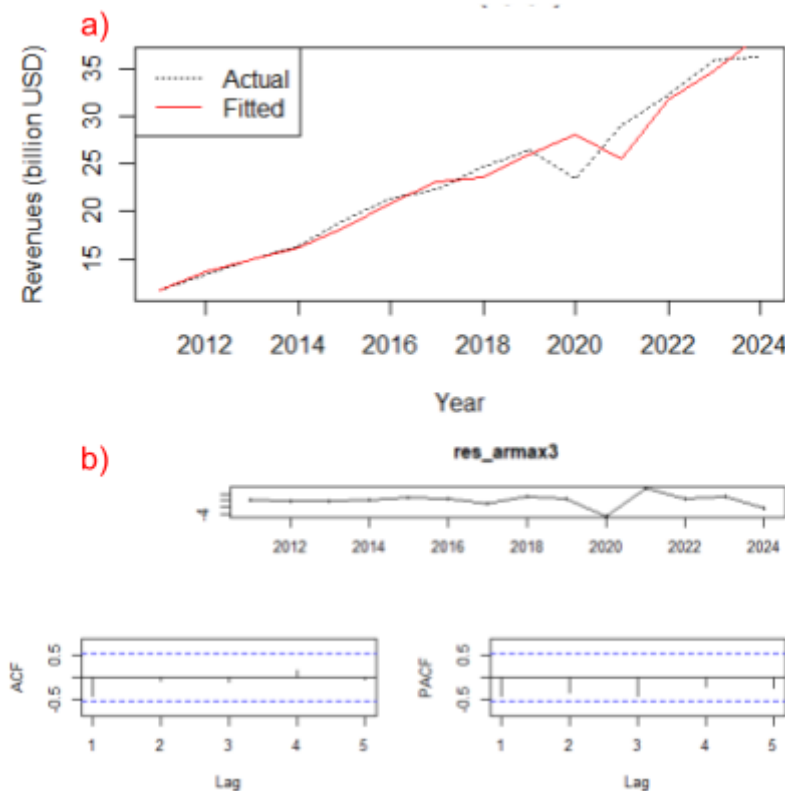


Figure 7. ARIMAX(0,1,0)'s plots of a) model's fit vs actual data, b) residuals, ACF and PACF

In Figure 7 it can be observed how the model properly fitted the values. However, the fitted values display poor responsiveness to short-term fluctuations. This behaviour suggests that the model's performance is mainly driven by the original dataset, and the exogenous advertising variable do not really affect its trajectory. This analysis is coherent with the mathematical results obtained from the t-test.

If we consider the residuals, we see that most of the pattern has disappeared, excluding the decline in 2020. However, the ACF and PACF do not suggest any significant lag.

B. Generalized additive model and Prophet model

In order to look for more flexibility, the generalized additive models are considered. The first try was based on the linear model, assuming a linear relationship between both predictors and response, so in practice a linear regression was implemented. The second trial consisted in adding a smoothing on the spendings, which gives a worse result that is proved by the fact that the flexible function for the spending predictor is not significant, suggesting that the relationship isn't strongly non-linear. For this reason a linear effect is sufficient and preferable.

Again based on the GAMs, for the prophet models we observed both linear and logistic growth, adding also an external variable which is the one referred to the advertising spendings in order to make a clear comparison between the special linear GAM and the

Table 6. Comparison of MSE for GAM models

MODEL	MSE
Linear GAM	1.2927
GAM with smoothing	0.9961
Prophet: linear growth	1.2921
Prophet: logistic growth	1.2667

prophet model with linear growth, even if considering residuals, ACF and PACF plots are pretty similar among the linear and logistic growth.

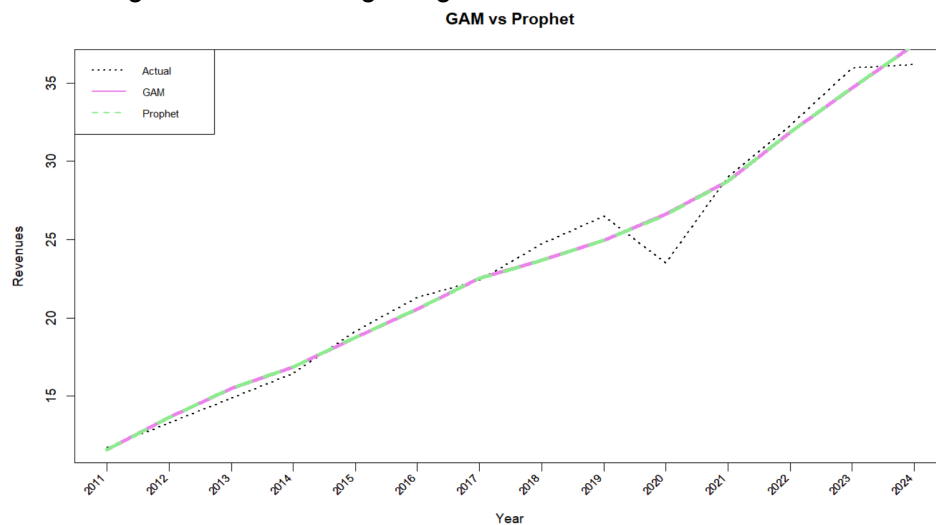


Figure 8. Plot comparison between prophet and GAM model.

It's visible from the comparison plot that the linear GAM and the prophet with linear growth are overlapping, given that the small dataset implies the implementation of non complex model and for the same reason since with few data points the metrics could be misleading, a parsimonious model like the GAM could be preferable.

Comparison of the best models

Since some of our models did not have an AIC value, we decided to implement MSE as an alternative to compare all our models. In the following *Table 7* and *Table 8*, it can be found a comparison of all the different types of models which displayed the best performance in their category. Further analysis of these results will be discussed in the following section.

Table 7. Comparison of best models' performance on quarterly data		
MODEL	AIC	MSE
TSLM (trend + seasonality)	99.9155	0.2264
GBM with exponential shock		0.1203
Auto.ARIMA(0,1,2)	86.9066	0.1963
SARIMA(1,1,1)(1,1,1)[4]	82.73706	0.137
ETS	142.3077	0.004
Prophet (linear growth) with holiday		0.1458

Table 8. Comparison of best models' performance on yearly data		
MODEL	AIC	MSE
auto.ARIMAX (0,1,0)	58.5971	3.1079
Linear GAM	51.3253	1.2977

Discussion

Based on the results previously presented, let's now analyze our initial hypotheses to determine their correctness. The first hypothesis was: "A linear regression model will only fit the trend and seasonality properly". This hypothesis was correct, as expected the TSLM failed to model the decline in sales during the pandemic, but it provided good results due to its ability to model trend and seasonality. Moreover, our second hypothesis was also supported by the results since the GBM with exponential shock was the only model that managed to properly fit the dip in the 2020 year. Even though it was the model with the second lowest MSE in the category of models applied in quarterly data, its residuals did not display a white noise behaviour, probably due to the trend and seasonality of the series.

When discussing our third hypothesis the analysis becomes more complex. We expected the ARIMA models to provide the best solution for this dataset, and this is only true when we consider AIC as our performance's measurement. However, when taking into account the MSE, the ETS turns out to be the "best" model. At first, this result might look contradictory, but when taking into account both AIC and MSE to compare the performance, it's evident that the ETS model is overfitting. This affirmation is supported by the extremely low MSE which is a direct sign that the model is memorizing the dataset; and, the considerably high AIC, which is a measure that better reflects the quality of the model by penalizing the extra parameters. For all these reasons, the SARIMA model was chosen as the most robust model by providing a coherent result in both AIC and MSE, and a great modeling of the data confirmed by its residuals and ACF and PACF plots.

When comparing the ARIMAX and GAM models, which were applied on the yearly dataset with the exogenous variable of advertising spendings, we obtained controversial results. For the ARIMAX model the external variable is not statistically significant, which means that the advertising expenditures do not affect Starbucks' revenues. The ARIMAX results suggest that, after controlling for autocorrelation and trend, the marginal linear effect of advertising is weak, therefore it does not improve the forecasting performance and so no new advantageous information is obtained.

On the other hand, the GAM analysis identified this variable as significant at the 5% level ($p = 0.026$) in the linear case. However, the significance vanishes when smoothing is introduced, which supports the assumption of a linear relationship. While this p-value indicates significance at the standard $\alpha = 0.05$ threshold, it is not highly robust, particularly as it diverges from the ARIMAX results.

The final model incorporating advertising spend is the Prophet model. Similar to the GAM results, Prophet produces a slightly curved trend and a low MSE. However, the estimated coefficient for the exogenous variable in the Prophet model is approximately 0.05. This suggests that while a relationship may exist, advertising provides only a marginal contribution to the overall model.

Moving to the remaining prophet models, we observe that concretely in the prophet models with quarterly data, both additive and multiplicative seasonalities were considered and while the first one showed in every case a higher MSE, the second one seemed to fit more accurately following also the level of the datapoints and some detailed oscillations. Conversely, even if the metric itself would appreciate a lower MSE, similarly to what occurred with ETS, it's important to check the residuals as a result like this might imply overfitting. This is the reason that at the end among the prophet models, the one which seems the finest was with the additive seasonality and exactly with the shock defined as holiday, guiding the model to consider its impact as isolated.

Lastly, when looking at each model and comparing it with the exogenous data, it is observable that as a whole, the advertising spend acts like a horizontal line in reference to the changing revenues. This is especially true after normalization. It is then understandable that there is no strong level of correlation being established, as there just is not enough movement in the exogenous variable of advertising spend to begin with. While some of the models might reflect a level of marginal significance, it is also observable that this level of change in advertising spend cannot and will not ultimately serve as a useful predictor.

Conclusion

One of our main findings is that the expenses of advertising seem to have a small positive effect on the revenues and it does not lead to better forecasts. Hence, advertising expenses may support revenues but it does not appear to play a key role in the revenue growth.

Moreover, fitting the dip presented in 2020 is not enough to obtain a good model, capturing it is important for fitting the data but it's a necessary and not sufficient condition. A robust model should capture the overall trend and seasonality, avoiding overfitting this specific event in order to prevent an apparent almost perfect performance on the training data and a poor generalization. To ensure the models' quality, residual diagnostics were examined checking if the shock had been sufficiently absorbed by the model and no other pattern remained unexplained.

Finally, when defining the ideal model for a specific task, it's very important to have in mind what the main objective of the research is. In our case, we focus on an explanatory variable to help us understand the patterns in Starbucks' revenues, and that's why we choose a SARIMA model as the perfect candidate. This model proved to be robust enough in terms of AIC, MSE, and its results were confirmed by the ACF, PACF and residuals plot.

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